# CareNova
## Hospital Networks

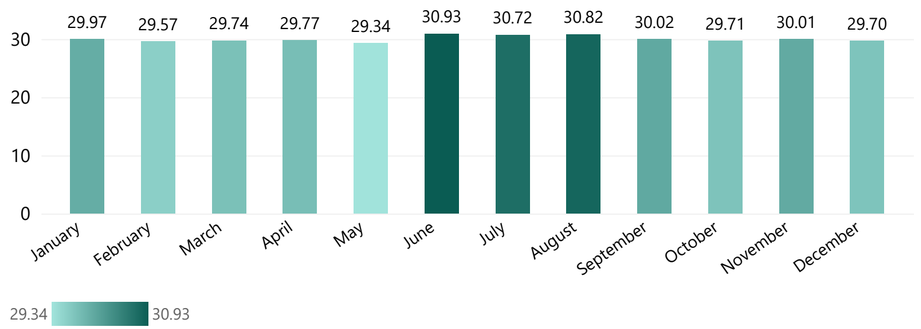
- Overview
- Clinical
- Financial
- Recommendations

# Hospital Overview

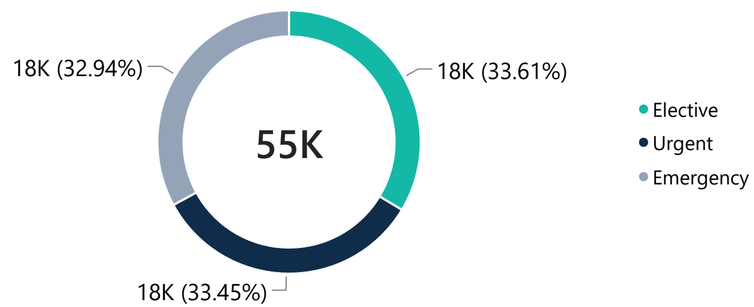
Year: All | Medical Condition: All | Insurance Provider: All

| Total Revenue | Total Admissions | Avg LOS(days) | Abnormal Tests % |
|---|---|---|---|
| 1.40bn | 54,860 | 15.50 | 33.54% |

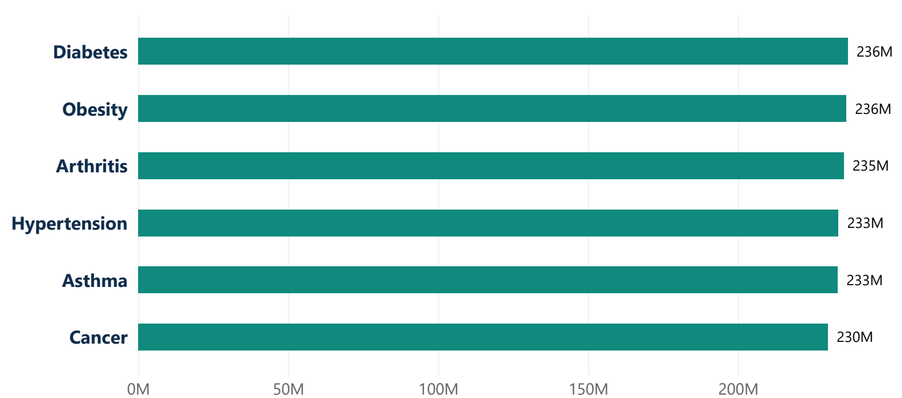
### Avg Daily Admissions by Month



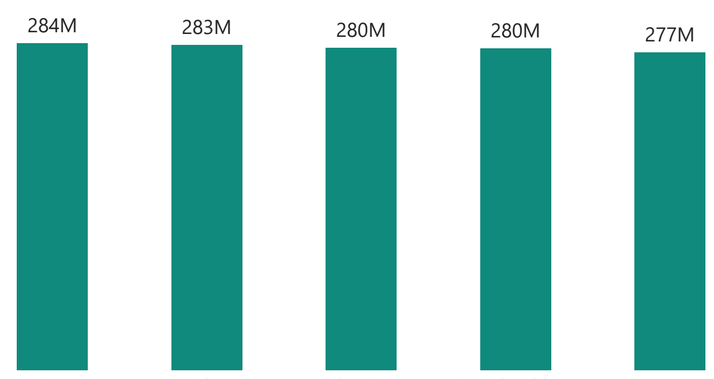

### Admissions by Type




### Revenue by Condition




### Revenue by Insurance Provider



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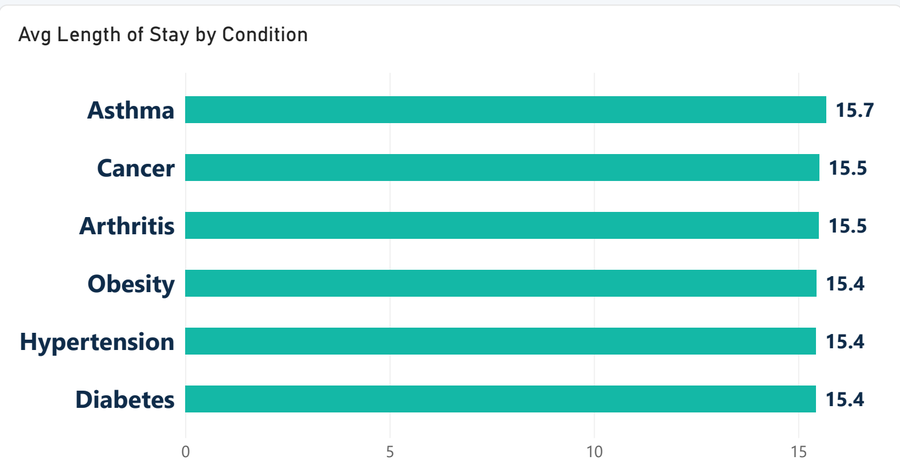
# Clinical Analysis

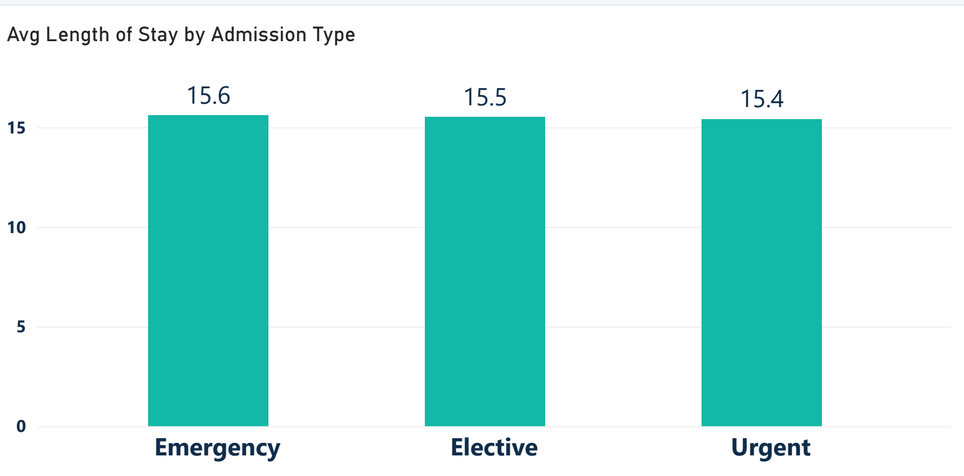
Year: All

Medical Condition: All

Insurance Provider: All

| Avg Patient Age | Emergency Admission % | Avg LOS(days) | Abnormal Tests % |
|---|---|---|---|
| 51.53 | 32.94% | 15.50 | 33.54% |

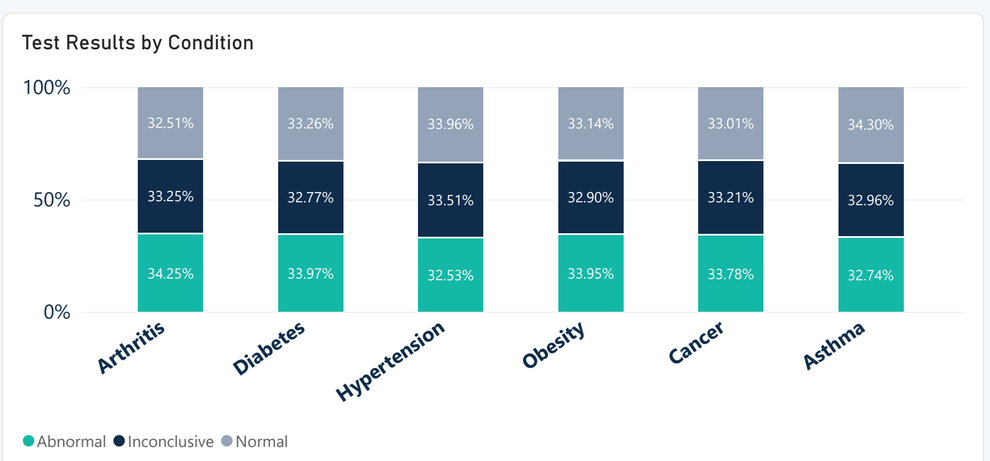



## Key Insights

- Length of stay is consistent across conditions (about 15.5 days). Asthma is highest at 15.7 days and Diabetes is lowest at 15.4.
- Admission type does not change length of stay: Emergency 15.6, Elective 15.5, Urgent 15.4 days.
- Test results are split almost equally (about one third each) in every condition, so no condition stands out for abnormal results.

**Suggestion:** since stay length is uniform, a hospital-wide discharge planning review is more likely to reduce average stay than condition-specific programs.



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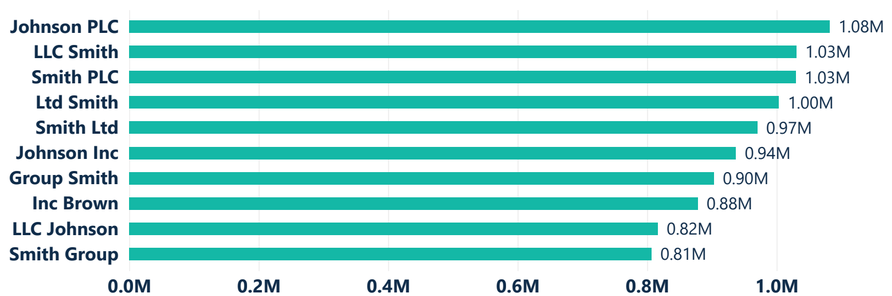
Recommendations

# Financial Analysis

Year: All | Medical Condition: All | Insurance Provider: All

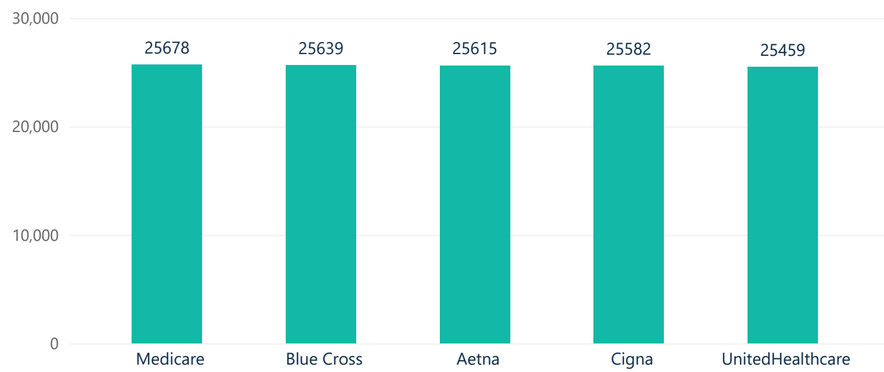
| Total Revenue | Total Admissions | Avg Billing per Admission | Avg Admission per day |
|---|---|---|---|
| 1.40bn | 54,860 | 25,594.63 | 30.03 |

Top 10 Hospitals by Revenue




**Note:** Hospital names are almost all unique in this dataset, so this ranking is for reference only.

Avg Billing per Admission by Insurance

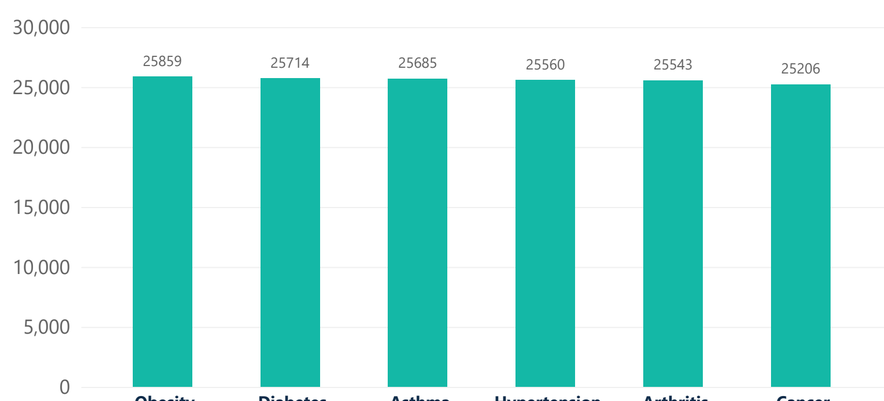



## Key Insights

- Average billing is about 25.6k per admission and varies very little by condition: Obesity is highest (25,859) and Cancer lowest (25,206), a gap of about 2.5%.
- Insurance providers look alike too: each brings about 20% of revenue (Cigna 20.25%, Aetna 19.69% ), and average billing differs by less than 1%.
- Revenue is not concentrated. The top hospital contributes only 0.08% of total revenue, and the hospital ranking is for reference only.

**Suggestion:** since billing and volume are evenly spread, revenue growth is more likely to come from increasing admissions than from targeting one condition or insurer.

Avg Billing per Admission by Condition



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# Insights & Recommendations

## Key Findings

1. Admissions are steady at about 30 per day all year, with a small summer peak in June to August (about 30.9 vs 29.3 in May).
2. Revenue is spread evenly across the six conditions: Diabetes is highest (236.5M) and Cancer lowest (229.9M), a gap of about 3%.
3. Average length of stay is 15.5 days, with no meaningful difference by condition or admission type (15.4 to 15.7).
4. The five insurers each bring about 20% of revenue (Cigna 20.25%, Aetna 19.69%).
5. Test results split into thirds in every condition (Normal 33.4%, Abnormal 33.5%, Inconclusive 33.1%).
6. Hospital and doctor names are almost all unique (39,876 hospitals, 40,341 doctors), so ranking them is not meaningful.

Data quality: 534 duplicate rows and 106 negative billing records were removed, leaving 54,860 clean admissions.

## Recommendations

**1. Plan summer capacity**
June to August is about 5% busier per day than the quietest month. Schedule staff leave and bed planning around this window.

**2. Run a hospital-wide discharge planning review**
Stay length is the same (about 15.5 days) for every condition, so no single condition drives it. Cutting one day would free about 6% of bed-days, which can be used for more admissions.

**3. Fix the master data at source**
Create unique hospital and doctor IDs, block duplicate records and reject negative billing entries. About 1% of rows in this extract needed removal, and hospital or doctor performance cannot be tracked without proper IDs.